

Executive Summary

Olist Brazilian E-Commerce Business Analysis

This project analyzes the **Olist Brazilian E-Commerce dataset** using SQL to evaluate business performance across customers, sales, products, logistics, and payment behavior. The objective was to answer key business questions and provide data-driven recommendations that support strategic decision-making.

The analysis covered **96K orders, 112,650 order items, 103,886 payment records, and 32,340 products** from **September 2016 to October 2018**. Advanced SQL techniques including **CTEs, subqueries, window functions, aggregations, and date functions** were used to complete ten business investigations.

Key Findings

- The **repeat customer rate was 3.12%**, indicating that customer retention is a major opportunity for improvement.
- **August 2018** recorded the highest monthly revenue and delivered orders, while September experienced a significant decline, warranting further investigation into seasonality and marketing activities.
- A small number of product categories generated the majority of revenue, whereas several categories consistently underperformed and require review.
- The highest-value customers were identified by **total revenue**, with many generating substantial revenue from a single high-value purchase rather than frequent purchases.
- Delivery analysis revealed that the largest delays occurred between **carrier pickup and customer delivery**, particularly in states such as **AM, PA, and PB**.
- **Credit cards accounted for approximately 78% of total revenue**, making them the dominant payment method, while debit cards showed lower adoption despite higher average order values than voucher payments.

Business Recommendations

- Improve customer retention through targeted loyalty programs and personalized marketing campaigns.
- Prioritize investment in consistently high-performing product categories while reviewing low-performing categories for optimization.
- Investigate logistics delays in underperforming regions and evaluate carrier performance to improve delivery efficiency.
- Continue optimizing the credit card payment experience while analyzing opportunities to increase adoption of alternative payment methods.

This project demonstrates the ability to transform raw transactional data into actionable business insights using SQL, supporting informed decision-making across marketing, operations, product management, finance, and customer strategy.